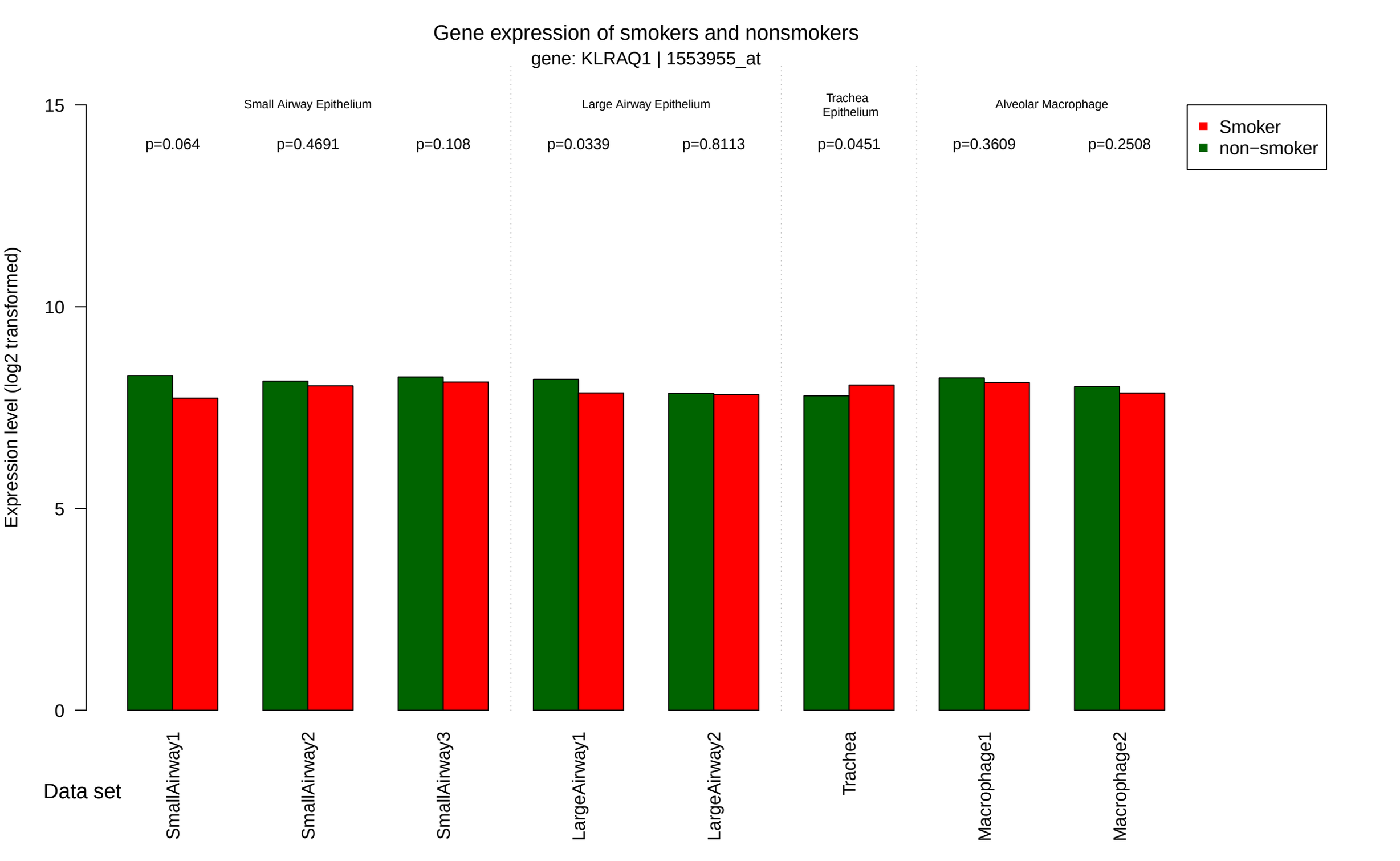
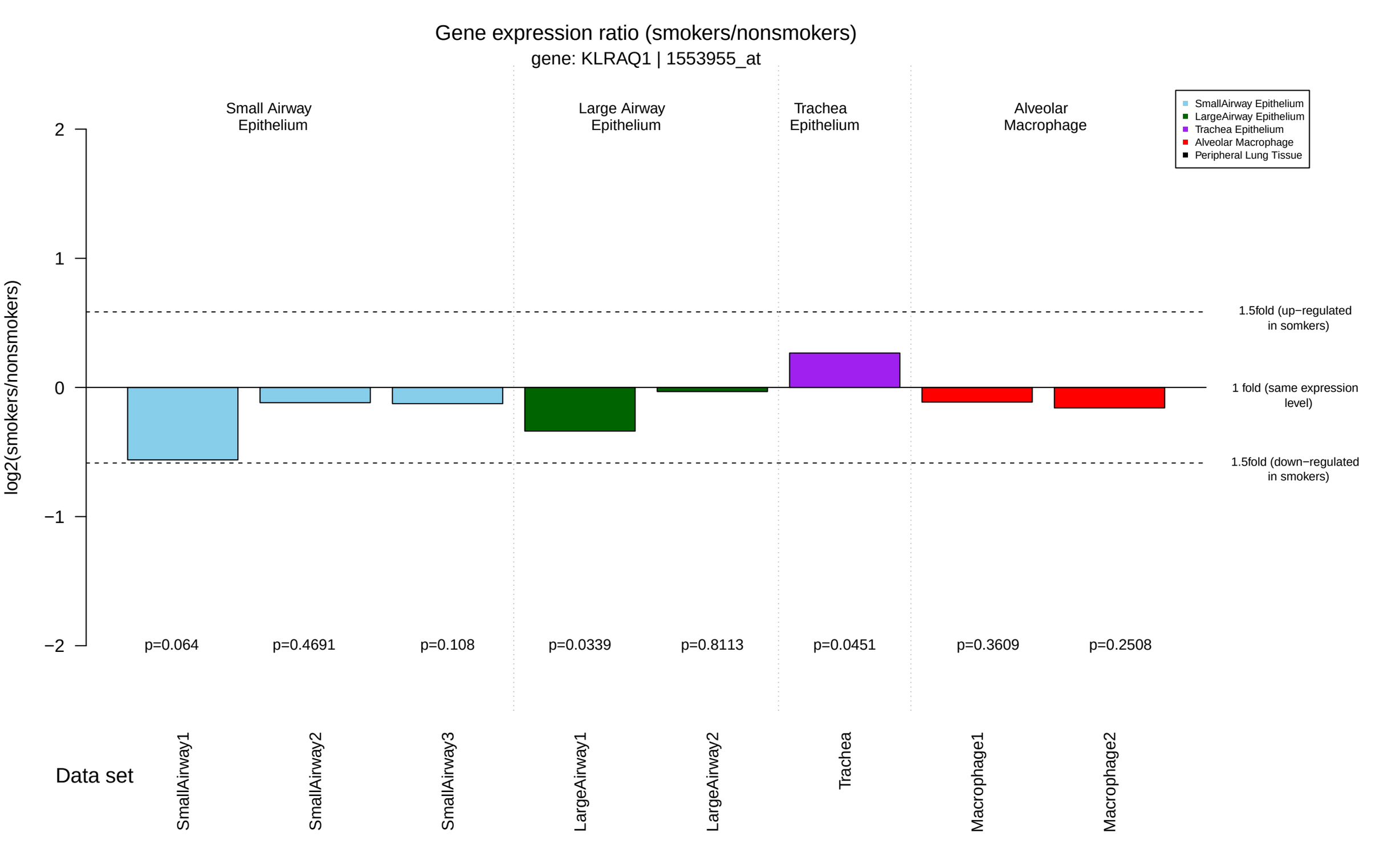
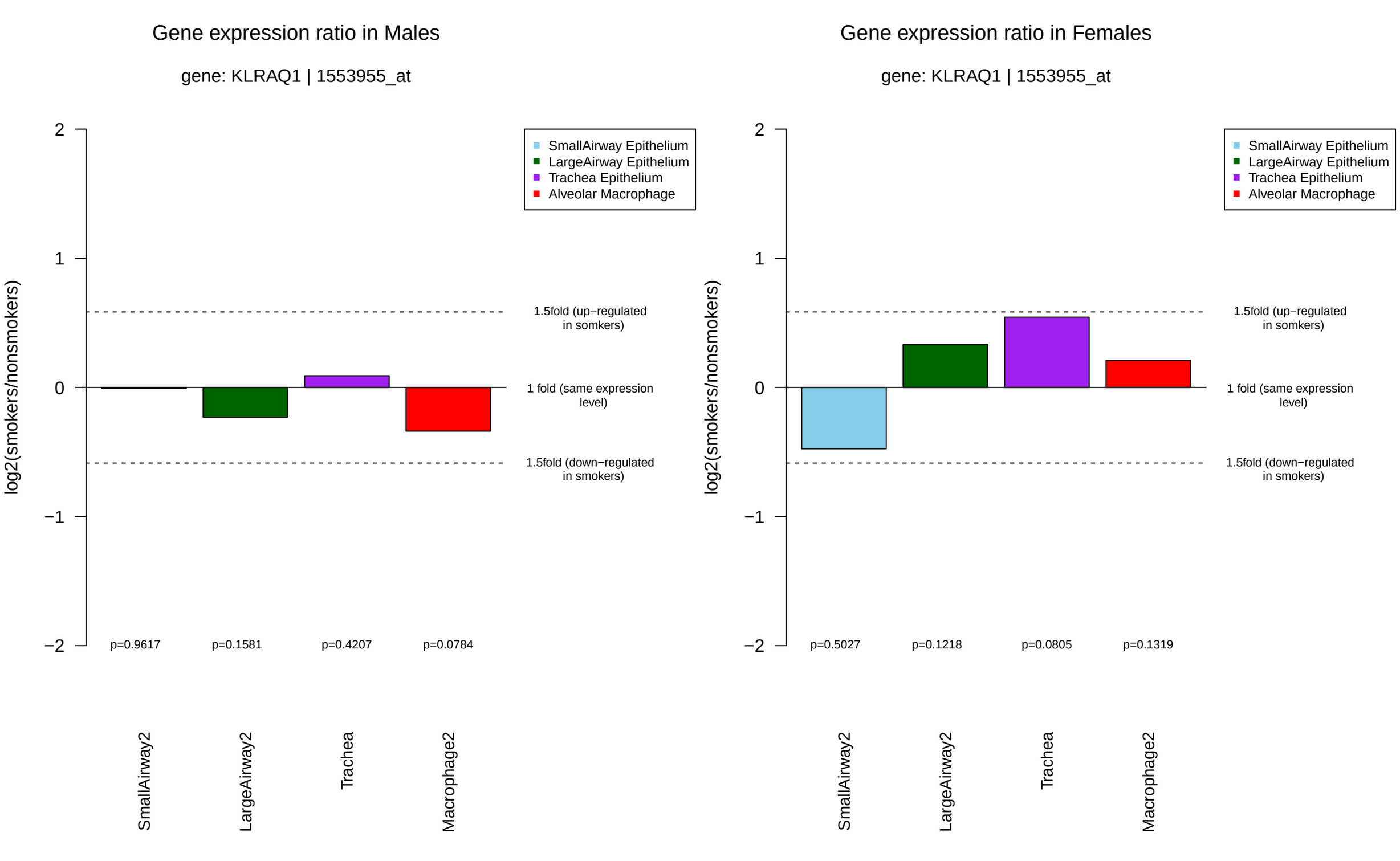
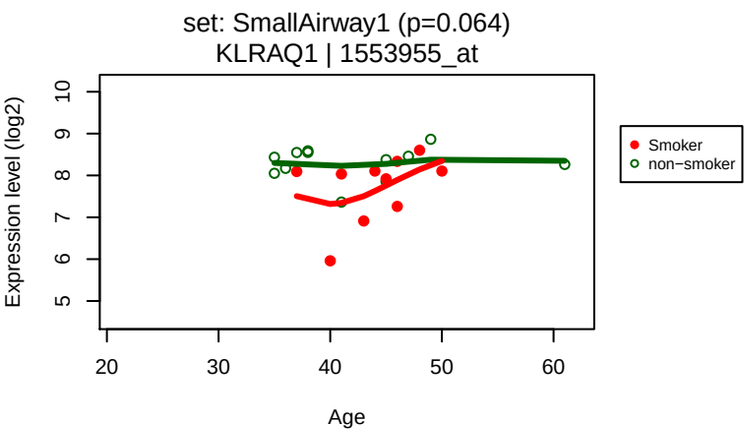
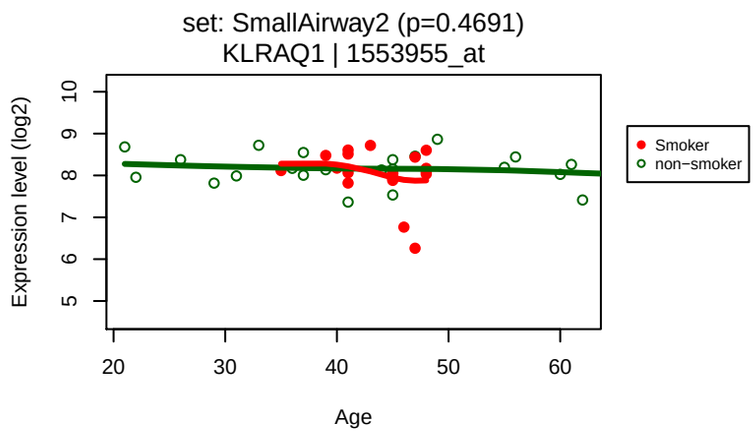
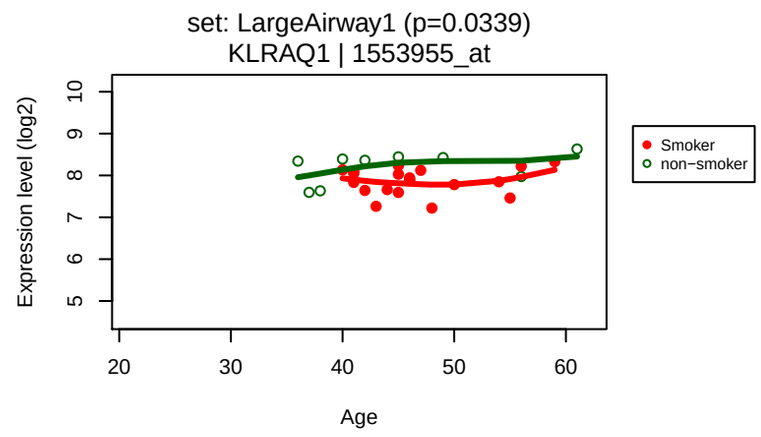

Gene expression of smokers and nonsmokers
gene: KLRAQ1 | 1553955_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.064
p=0.4691
p=0.108
p=0.0339
p=0.8113
p=0.0451
p=0.3609
p=0.2508
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

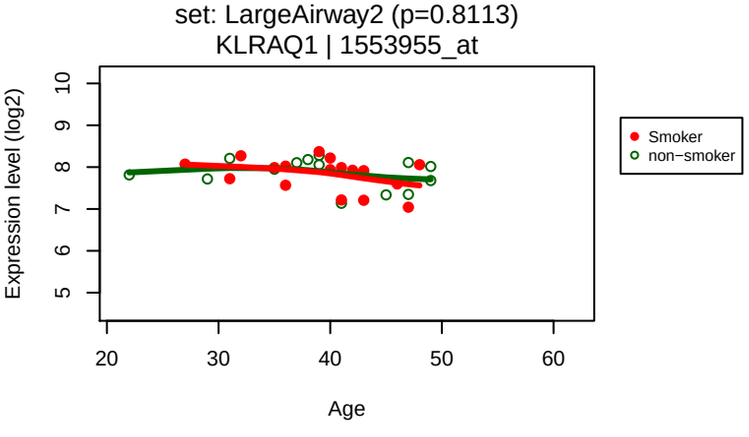

Gene expression ratio (smokers/nonsmokers)
gene: KLRAQ1 | 1553955_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.064
p=0.4691
p=0.108
p=0.0339
p=0.8113
p=0.0451
p=0.3609
p=0.2508
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

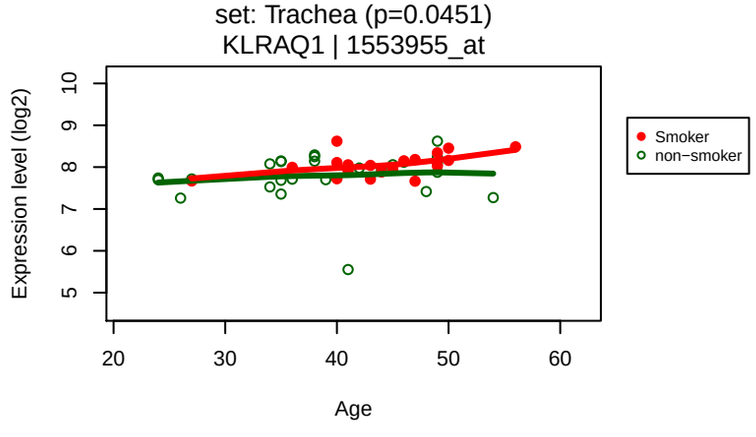

Gene expression ratio in Males
gene: KLRAQ1 | 1553955_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9617
p=0.1581
p=0.4207
p=0.0784
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLRAQ1 | 1553955_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5027
p=0.1218
p=0.0805
p=0.1319
SmallAirway2
LargeAirway2
Trachea
Macrophage2

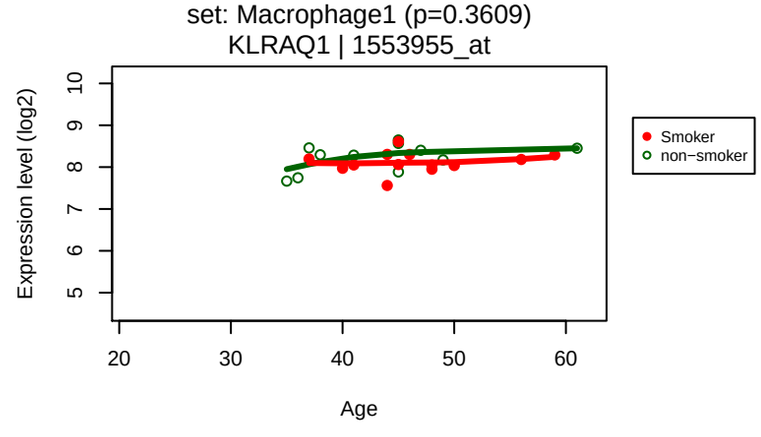

set: SmallAirway1 (p=0.064)
KLRAQ1 | 1553955_at
Expression level (log2)
Age
Smoker
non−smoker

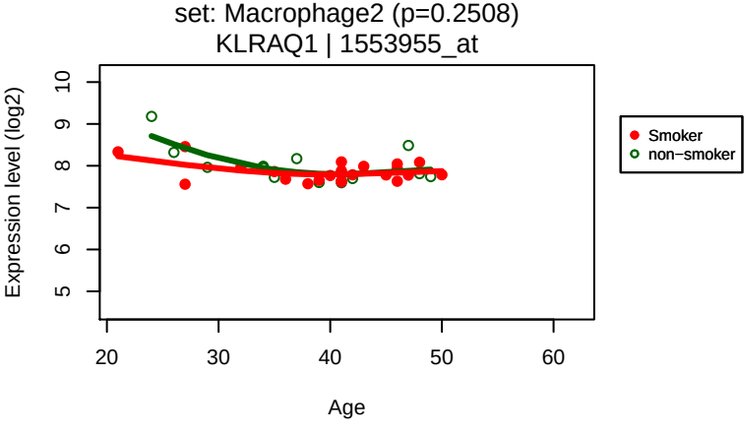

set: SmallAirway2 (p=0.4691)
KLRAQ1 | 1553955_at
Expression level (log2)
Age
Smoker
non−smoker

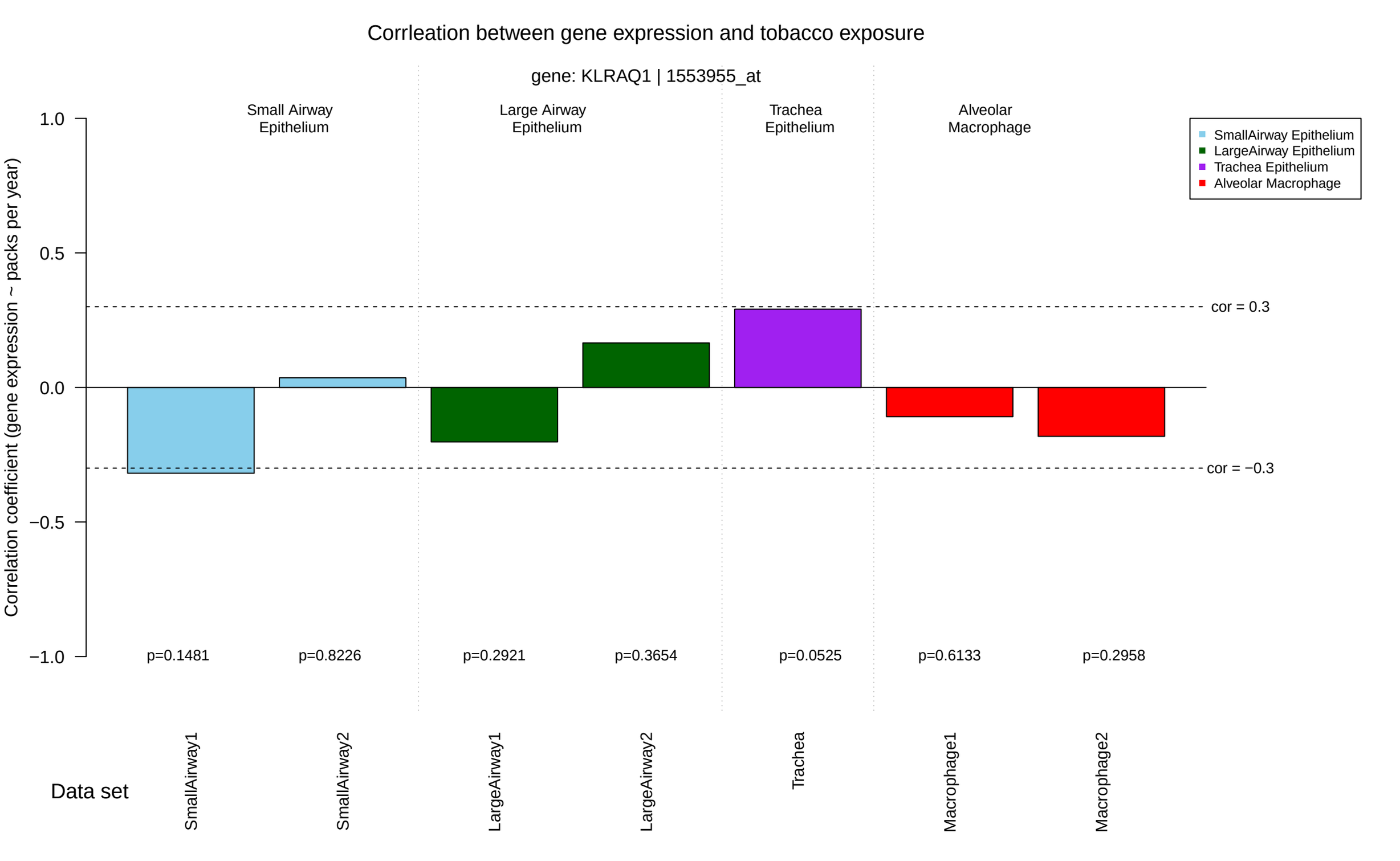

set: LargeAirway1 (p=0.0339)
KLRAQ1 | 1553955_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8113)
KLRAQ1 | 1553955_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0451)
KLRAQ1 | 1553955_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3609)
KLRAQ1 | 1553955_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2508)
KLRAQ1 | 1553955_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLRAQ1 | 1553955_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1481
p=0.8226
p=0.2921
p=0.3654
p=0.0525
p=0.6133
p=0.2958
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2